

Calibration of Production Transient Alert Brokers: Preparing Decision Infrastructure for LSST

Can broker confidence scores be trusted for automated follow-up decisions?

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QR code

Code, tables & figures
branch [alerce-selective-deferral](#)

1 Motivation

“When a broker reports 80% confidence, is it actually correct 80% of the time?”

Millions of alerts → Broker probabilities → Gates & rankings →

Limited follow-up

- Rubin/LSST alert volumes cannot be inspected manually.
- A classifier can rank the correct class first while still reporting an unreliable confidence.

Underconfidence

Correct predictions score too low → useful objects fail fixed gates.

Overconfidence

Wrong predictions score too high → unreliable labels auto-accepted.

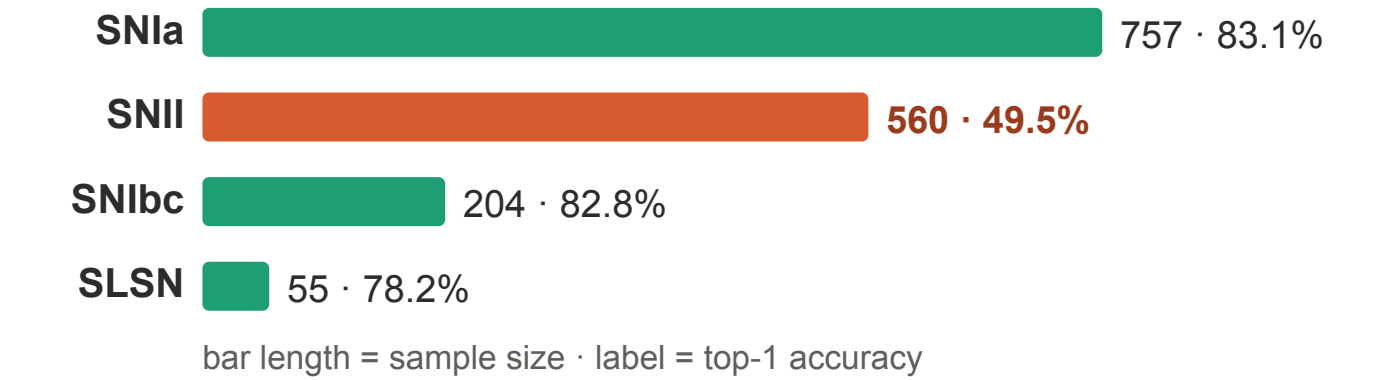
A functioning classifier is not automatically functioning decision infrastructure.

2 Data & ground truth

Spectroscopically confirmed ZTF Bright Transient Survey objects, cross-matched against the deployed outputs of two production classification systems. BTS spectroscopy is the independent ground truth.

System	Role	Task
ALeRCE light-curve RF	production alert broker	15-class
NEEDLE	rare-transient classifier (not a broker)	3-class

ALeRCE class counts and accuracy



Overall ALeRCE accuracy 70.9%, but strongly class-dependent — **SN II is the classification bottleneck**.

PROPOSED Calibration-aware spectroscopic deferral

Broker probabilities
↓
Calibration layer
↓
Checker risk $r_i = P(\text{classification is wrong})$
↓
Scientific value & observability
↓

Top-B targets for spectroscopy

$$\max_z \sum_i z_i t_i V_i q_i \quad \text{s.t.} \quad \sum_i z_i c_i \leq B$$

z_i select for spectroscopy · r_i checker error risk · V_i scientific value · q_i probability of a usable spectrum · c_i exposure cost · B observing budget. In words: **choose the candidates where a spectrum buys the most science within the available budget**. A separate novelty channel will flag objects outside the broker taxonomy.

Not implemented or tested here — these results *motivate* a future calibration-aware learning-to-defer framework.

3 Method

1 **Score correctness.** Top-ranked class vs independent spectroscopic label.

2 **Audit calibration.** Does reported confidence match observed accuracy?

$$\text{ECE} = \sum_b (n_b/N) \cdot | \text{acc}(b) - \text{conf}(b) |$$

3 **Recalibrate.** Held-out temperature scaling. Brokers expose probabilities, so log-probabilities act as pseudo-logits and softmax returns a valid simplex.

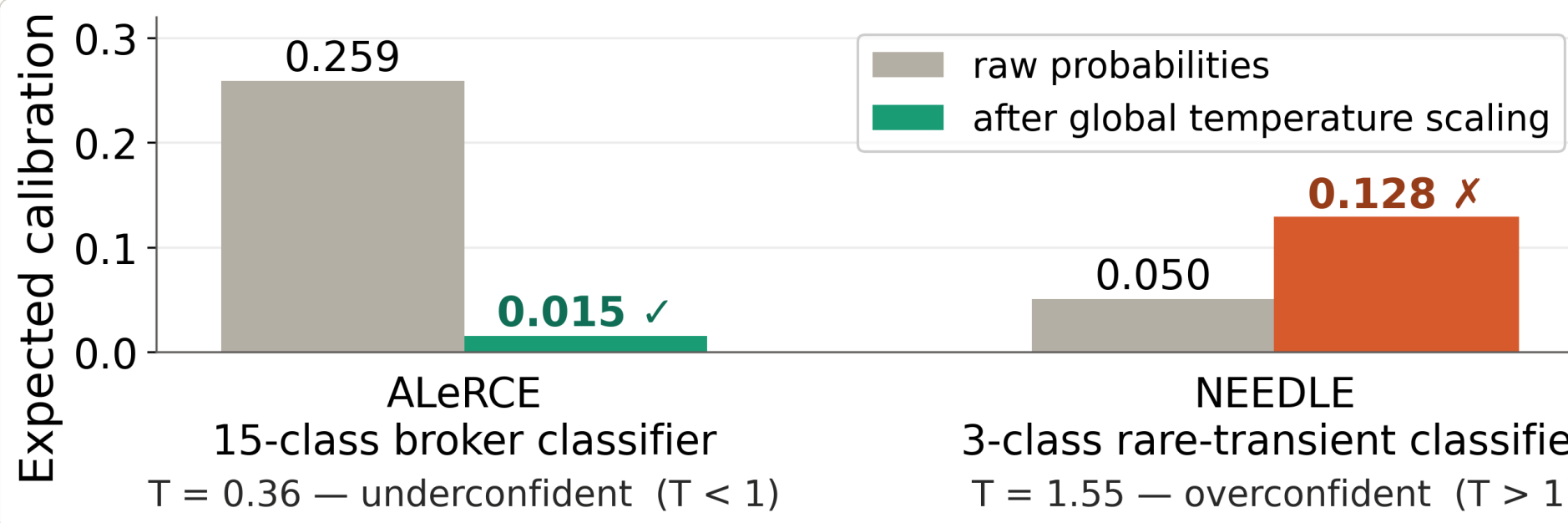
$$\tilde{\mathbf{p}} = \text{softmax}(\log \mathbf{p} / T) \quad \text{monotonic} \rightarrow \text{top-1 class unchanged}$$

4 **Train a reliability checker.** Logistic model estimating **P(top-1 correct)** from the calibrated vector plus its shape (margin, entropy, branch mass), benchmarked against a predicted-class-rate control.

5 **Accept or defer.** Rank by estimated reliability; defer the least trustworthy fraction.

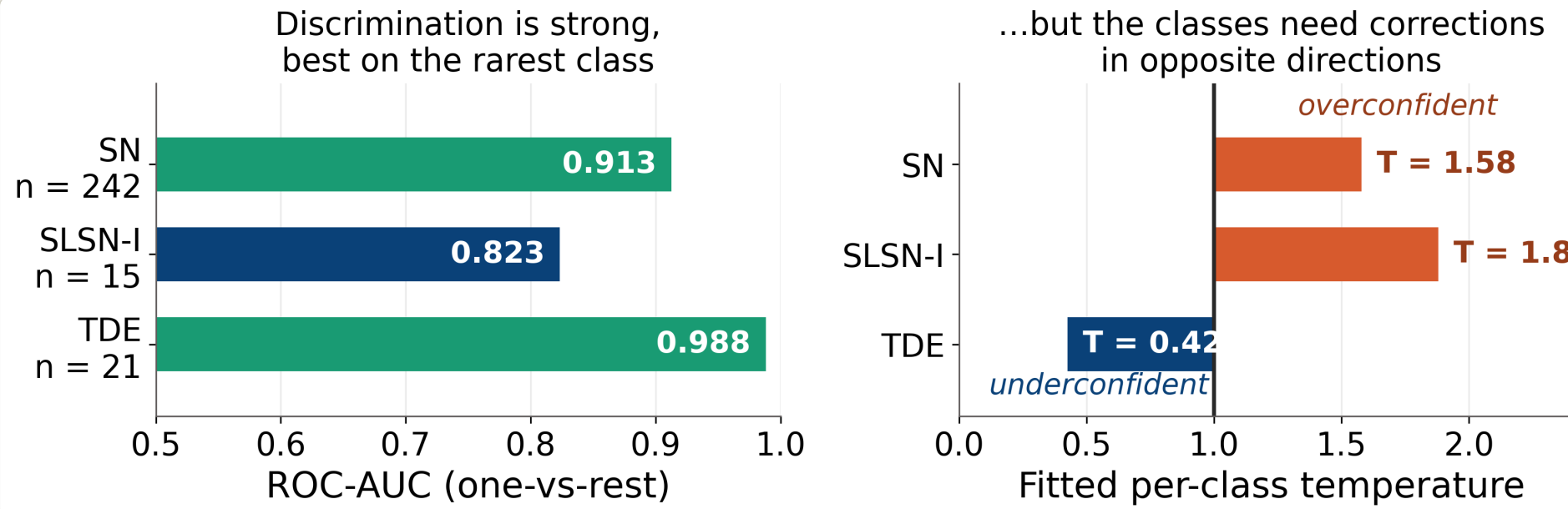
Five-fold out-of-fold · paired stratified bootstrap · ROC-AUC, AURC and risk-coverage. *Deferral means more photometry or human review — not automatically spectroscopy.*

4 Two systems, opposite failure modes



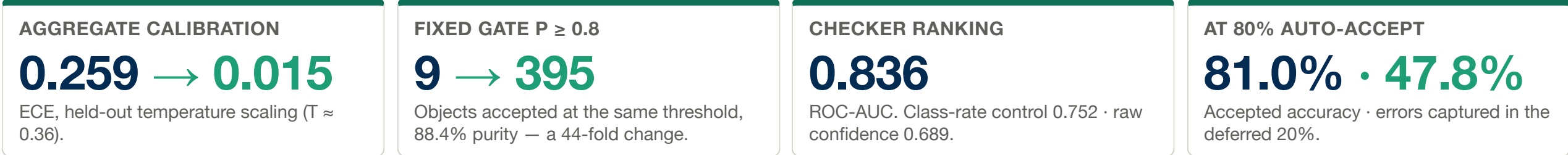
ALeRCE is underconfident and is repaired. **NEEDLE** starts far better in aggregate but is overconfident, so the identical procedure **makes it worse**. Aggregate ECE does not tell you which correction a system needs.

NEEDLE per class: strong ranking, conflicting calibration

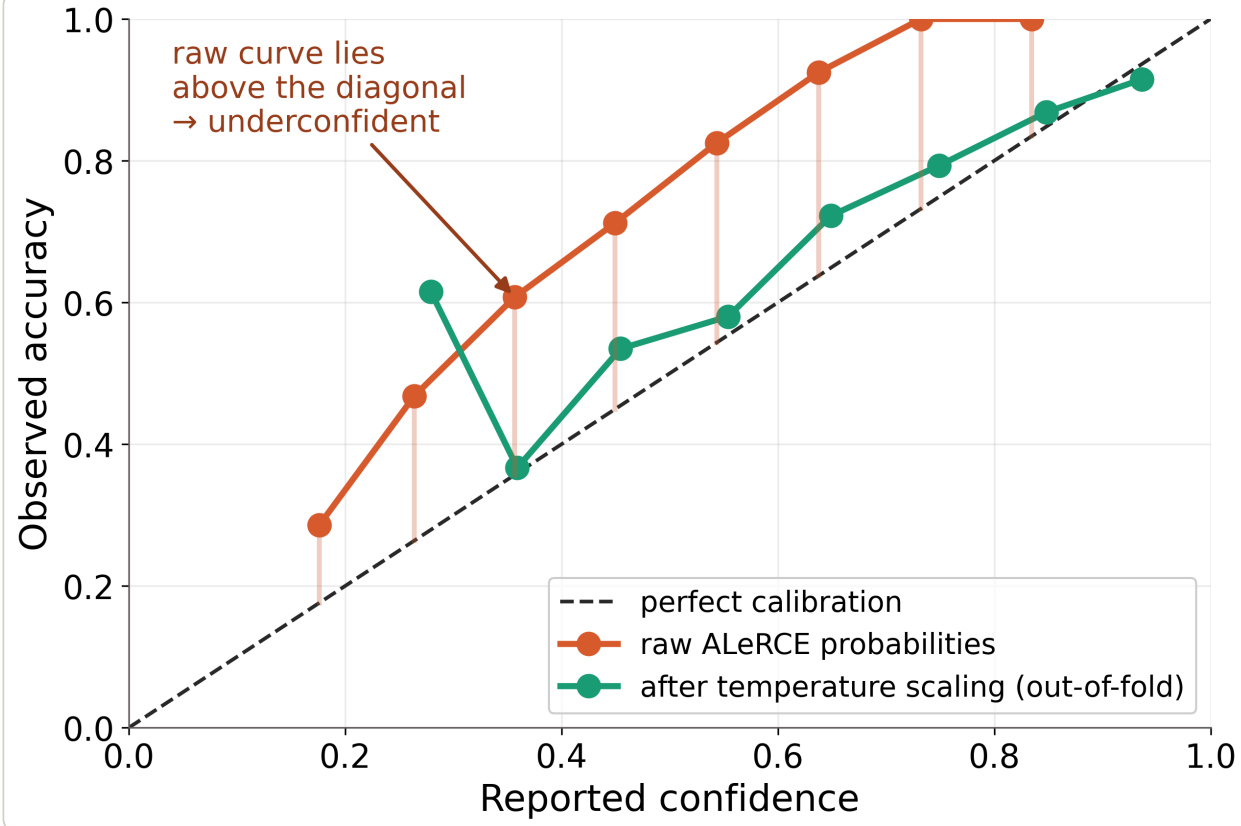


NEEDLE ranks its classes well — best on its rarest, **TDE (ROC-AUC 0.988, 21/21 recovered)**. But its classes need corrections in **opposite directions**, so no single temperature serves all three — which is why the global fix backfires. *Rare-class samples are small: n = 15, 21.*

5 Results — repairing and using the ALeRCE scale

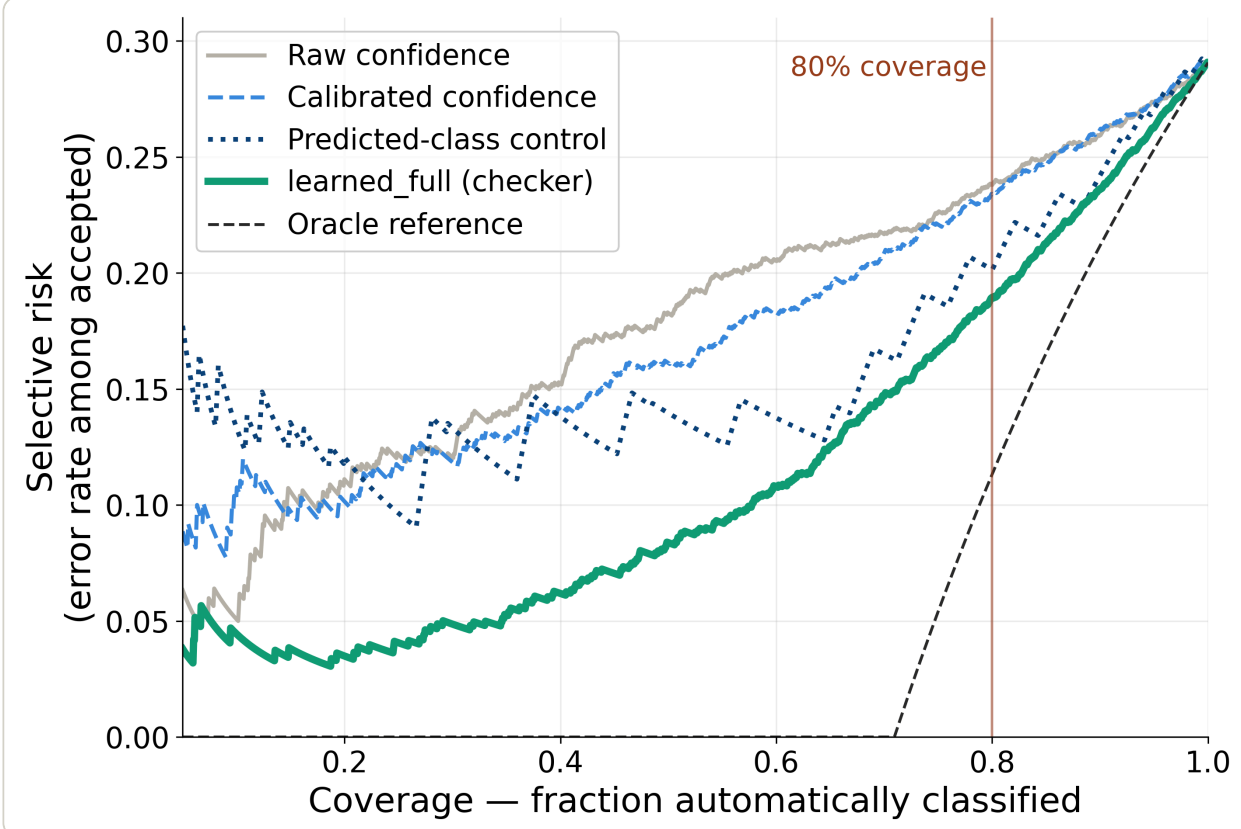


ALeRCE is systematically underconfident



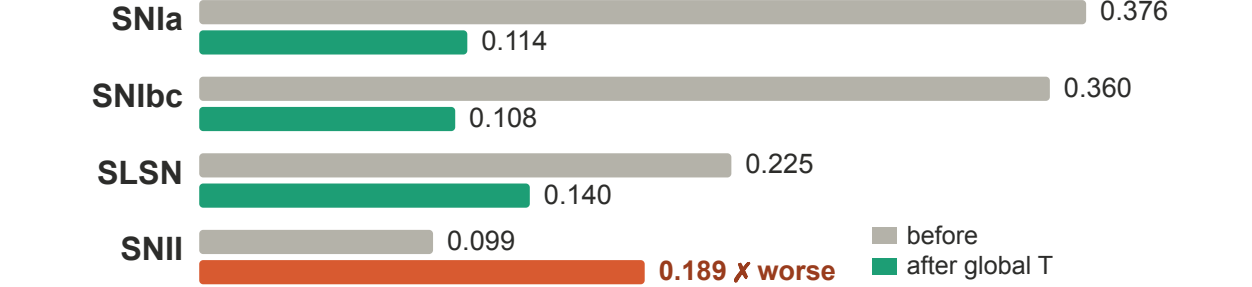
Raw ALeRCE sits far above the diagonal — correct far more often than it claims. Temperature scaling pulls it onto the diagonal *without changing any predicted label*.

Selective classification



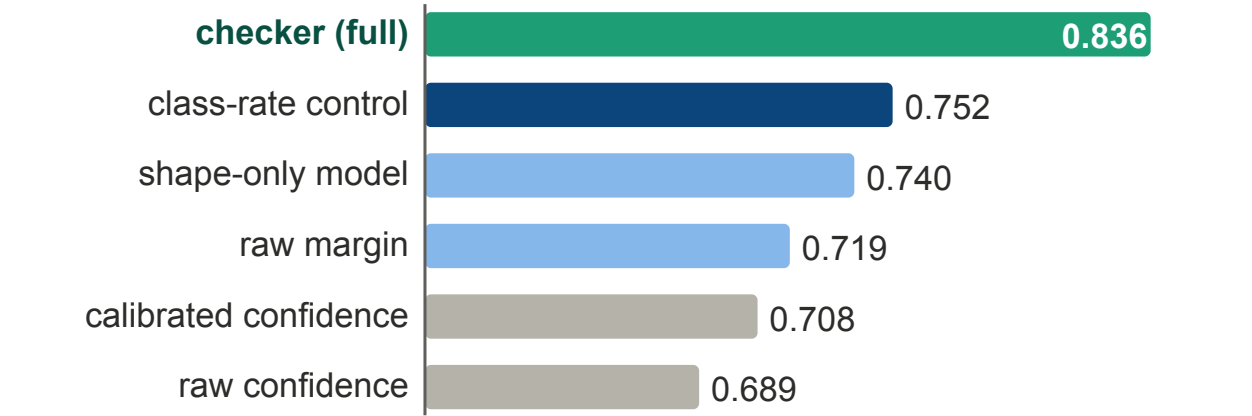
Policy @ 80% coverage	Acc.	Errors caught
Raw confidence	76.2%	34.5%
Predicted-class control	79.9%	44.5%
Checker (full vector)	81.0%	47.8%

...and not uniformly across its classes



Per-class ALeRCE ECE. SN Ia and SN Ibc are repaired; **SN II degrades** — the aggregate improves while one class gets worse.

The checker beats class identity alone



$\Delta = +0.083$ over the class-rate control, paired-bootstrap 95% CI [0.057, 0.107] — object-level reliability beyond class identity. The shape-only model does *not* clear the control: the signal is in the full 15-class vector.

6 Conclusions

- Deployed systems miscalibrate in *opposite* directions — ALeRCE strongly underconfident, NEEDLE overconfident — so the correction a system needs cannot be read off its aggregate ECE.
- Calibration repairs what a probability *means*. It never changes the predicted class, but it decides which objects clear a fixed gate.
- One global correction is not enough: it degrades ALeRCE's SN II, and NEEDLE's own classes need temperatures on opposite sides of 1.
- A checker reading the full probability vector identifies unreliable predictions beyond predicted-class identity, concentrating errors into a small deferred queue.

Calibration fixes the scale. A **reliability checker** decides which predictions to trust. Neither is optional if broker outputs are to drive automated follow-up.

A functioning alert classifier is not automatically functioning decision infrastructure.

Main references

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